Introduction to Econometrics

Pre-sessional Lecture

Reading

This lecture covers material from Wooldridge (2025):

- Chapter 1 - "The Nature of Econometrics and Economic Data"

Econometrics

Econometrics? What is it?

> "Econometrics is based upon the development of statistical methods for estimating economic relationships, testing economic theories, and evaluating and implementing government and business policy." (Wooldridge 2025, 2)

There are a few key elements here:

- statistical methods
- economic relationships
- testing theory
- policy evaluation

Purpose of Econometrics

Econometrics encompasses a set of tools used by Economists to empirically verify claims made about "the real world" (as observed in the **data**):

Purpose 1: Test economic theory.

e.g., **Efficiency wage theory:** paying workers in excess of their marginal product will increase effort (improve retention or reduce shirking).

How would we test this theory?

- Need data on worker wages and a measure of effort (?)
- Ideally, we want to compare similar workers, doing the same job, but being paid different wages.

The impact of economic policy is often theoretically ambiguous,

e.g., according to standard economic models, **minimum wages** should

- decrease in employment under perfect competition
- increase in employment under monopsony

Purpose 2: Evaluate the impact of economic policy.

- Statistical significance: Can we reject a null (zero) effect?
- Sign: Do we find a positive or negative impact?
- Magnitude: How big was the effect (on average)? [1]

> **Note**
>
> Evidence on the impact of a policy can be a used as a (in)direct test of economic theory. For example, the literature on Minimum Wages - which broadly finds no disemployment effect - is cited as evidence of monopsony power in the labour market.

Interjection: green boxes

You will notice in the notes that there is a green box!

> **Reduced-form vs Structural Estimation**
>
> Econometrics also encompasses the estimation of structural economic models. For example, estimating the macroeconomic Real Business Cycle model. In this setting, the goal is to estimate the structural parameters of the model, which then allows you to do policy evaluation and/or counterfactual analysis.
> This module will largely deal with the estimation of reduced-form models. These are models which focus on the relationship between economic variables, but do not necessarily identify structural parameters from an economic model.

- Non-examinable material

[1] Linked to the idea of economic significance and used in cost-benefit analysis.

- Extra information **for the curious**
- All other boxes ARE examinable
- Will not appear in future lecture slides

Data

A lot is going to depend on the nature of the data.

1. Cross-sectional data

    - Each observation represents a different unit (individual, firm, state, municipality,...), observed at the same period of time
    - e.g. British Social Attitudes Survey
    - Notation: y_i

2. Time series data

    - Each observation represents the same unit, observed over multiple periods of time
    - e.g. UK CPI
    - Notation: y_t

More complex complex data can include both a cross-sectional and time dimension

3. Repeated/pooled cross-section data

    - A collection of multiple cross-sections
    - e.g. British Social Attitudes for 1983-2025
    - Notation: y_{it}

4. Longitudinal/panel data

    - Multiple units (individual, firm, state) are observed over multiple periods of time
    - e.g. UK Household Longitudinal Study ("Understanding Society")
    - Notation: y_{it}

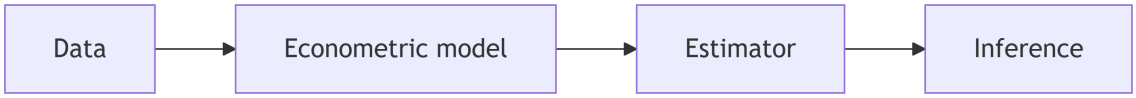
DGP

Datasets vary in another important dimension: the *data generating process* (DGP)

1. Experimental data
    - Key variables are the outcome of a randomized experiment (i.e. probabilistic)
    - *Controlled* experiments: researcher determines the assignment mechanism
    - Field (i.e. RCTs) and lab experiments are increasingly common in Economics
2. Observational (non-experimental) data
    - The true data generating process (or assignment mechanism) is unknown
    - Also referred to as retrospective data; e.g. surveys, administrative records, etc.

> **!** Natural Experiments
>
> *Natural* experiments occur when the assignment mechanism is outside the control of the researcher, but still creates the conditions of an experiment (i.e. treated and control). This assignment need not be random, but must be exogenous to factors relevant to the outcome. These are considered observational studies.

Econometric analysis



- Inference will depend on the properties of the estimator
- The properties of the estimator will, in turn, depend on the assumptions of the model
- The assumptions of the model have to fit (within reason) the data generating process.

> **i** Note
>
> This is a not a description of the research process, which typically begins with a research question and testable hypothesis. In reality, applied economic research questions are often constrained by the nature of the data and econometric model. This creates a feedback loop between the econometric analysis and research question. This is particularly the case in observational studies where the researcher does not know the data generating process (or control assignment in the experiment).

Causal Inference

When testing economic theory or evaluating policy, it is important to be able to separate out causation from correlation.

For example, to test efficiency wage theory

$$E[e_i|w_i = high] - E[e_i|w_i = low]$$

Experiments solve this problem through random assignment

- Higher wages are paid to a random subset of workers.
- In Lecture 6, we will discuss the Potential Outcomes Framework that underpins this theory.

Causal Inference

In observational studies, we can't rely on random assignment. Instead,

- Control variables: adjust for confounding factors
- Instruments: exploit exogenous sources of variation
- Transformations: transform the model to remove sources of bias

Ceteris Paribus

For example, to test efficiency wage theory, we might compare

$$E[e_i|w_i = high, c] - E[e_i|w_i = low, c]$$

where c is a set of worker characteristics:

- e.g. $c = \{education_i, age_i, gender_i, ethnicity_i\}$

Now the comparison is between two "similar" groups of workers.

> **Ceteris Paribus**
>
> A Latin phrase meaning "all other things being equal". A phrase that is used to infer causation from a comparison of similar, but not equal, groups.

Example

Brown (2011)

Paper: brown2011quitters "Quitters Never Win: The (Adverse) Incentive Effects of Competing with Superstars"

Theory: "Superstar Effect":

in tournaments,

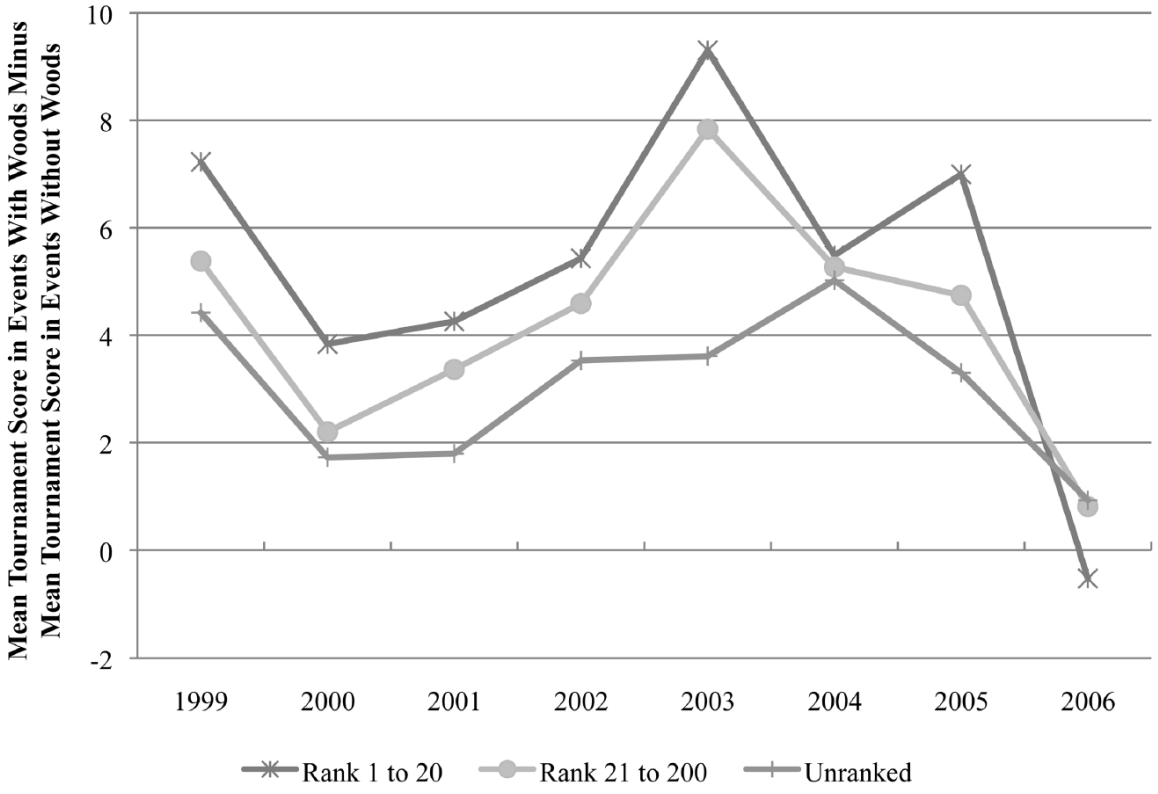
- with unequal distribution of talent,
- may be optimal for less talented to "give up" (reduce effort) when competing against a more talented individual

Data

> "Professional golf tournaments, where effort relates relatively directly to performance, present an opportunity to examine empirically the influence of a superstar." (pp. 983)



Unconditional differences



But these are different tournaments with potentially different players!

Econometric model

The author estimates a model that controls for certain player/event characteristics:

$$\begin{aligned} strokes_{ij} =& \beta_1 star_j \times HRanked_i + \beta_2 star_j \times LRranked_i \\ &+ \beta_3 star_j \times URanked_i + \alpha_1 HRanked_i + \alpha_2 LRranked_i \\ &+ \gamma_0 + \gamma_1 X_i + \gamma_2 Y_j + \varepsilon_{ij} \end{aligned}$$

where

- HRanked = top 20; LRranked = 21-200; URanked = unranked
- X_i: player characteristics
- Y_j: event controls (major dummy, weather, purse, field quality, viewership)

Estimation & Inference

TABLE 4
Regression Results for Woods's Unexpected Absences: Strokes Relative to Par

	Knee Surgery in 2008: July–September 2007–9		Personal Issues in 2009 and 2010: January–March 2009–10	
	First Round (1)	Tournament (2)	First Round (3)	Tournament (4)
Superstar effect for players:				
Ranked 1–20	.244	3.919***	1.125	2.754
	(.614)	(1.462)	(.930)	(2.038)
Ranked 21–200	.317	5.647***	1.510***	3.519***
	(.309)	(.956)	(.447)	(1.050)
Unranked	.119	5.755***	1.210***	3.554***
	(.377)	(1.094)	(.484)	(1.351)
Observations	2,676	1,616	2,339	1,172
Adjusted R^2	.21	.58	.18	.36

EC3301 Housekeeping

Syllabus

Week	Topic
1	Linear regression model
2	Ordinary least squares
3	Properties of OLS and inference
4	Hypothesis testing and heteroskedasticity (recorded)
5	Large sample properties of OLS
6	Independent Learning Week
7	Dummy variables and potential outcomes framework
8	Model misspecfication and omitted variables
9	Proxy variables and instrumental variables
10	Estimation with instrumental variables (recorded)
11	Simple panel data models and difference in differences

Key dates

- Week 4: Class Test 1
  - 50 minutes, 25% weight
  - Covering Lectures 1-3 (and Tutorial 1)
- Week 10: Class Test 2
  - 50 minutes, 25% weight
  - Covering Lectures 1-9 (with emphasis on 4-8)
- Week : Project
  - Individual, 50% weight
  - Due

Support

- Econometric Labs
  - 5x 2 hours
  - Weeks 2,4,8,9,10
- Tutorials
  - 4x 1 hour
  - Weeks 3,5,7,11
- Textbook
  - Wooldridge (2025) *Introductory Econometrics: A Modern Approach*, Eighth Edition, Cengage International Edition
  - **You can also use an earlier edition.**
- Office Hours
  - Tues & Wed 11:00-12:00, G3 Castlecliffe
  - No reservations required

Week	Lecture	Classes	Assessment

Overview

Week	Lecture	Classes	Assessment
1	Lecture 1		
2	Lecture 2	Lab 1	
3	Lecture 3	Tutorial 1	
4	Lecture 4 (recorded)	Lab 2	Class Test 1
5	Lecture 5	Tutorial 2	
6			
7	Lecture 6	Tutorial 3	
8	Lecture 7	Lab 3	
9	Lecture 8	Lab 4	
10	Lecture 9 (recorded)	Lab 5	Class Test 2
11	Lecture 10	Tutorial 4	

Contact

Moodle Forum: *Please use the Moodle forums to ask questions that may be of benefit to other students.*

- Culture of **supportive** online participation.

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Office: G3 Castlecliffe

MS Teams

Please do not contact me directly via MS Teams. Use the Moodle forums or email.



Code

Lecture material will contain parallel examples in Stata and R for all *relevant* code

Stata

```
display "Hello EC3301!"
```

```
Hello EC3301!
```

R

```
print("Hello EC3301!")
```

```
[1] "Hello EC3301!"
```

Stata and R?

This is a bit of an experiment: and you are the 'guinea pigs'

- Labs will be 'run' in Stata:
  - majority of students learnt Stata in EC2203;
  - tutors are primarily trained in Stata (as are most economists);
  - **BUT** labs are largely self directed;
  - you can choose to complete the R lab (during the lab)
- Tutorial material *may* contain some Stata output, only
- Why add R?
  - Continuity of learning: some students have worked with R in during sub-honours
  - Assessment: you will be allowed to use R for the project
  - Learning Outcomes: econometric analysis $\neq$ programming

Stata or R?



- easy to learn and use
- intuitive GUI
- `.dta` data format
- simple command-based 'language'
- programming *light*



- flexible language
- advanced programming
- industry standard (for data science)
- open source = FREE
- packages, packages, and *more* packages

(See Lab 1 for more information.)

Survival guide

Thriving in 3301

1. **Attend class:** lectures, labs, tutorials
2. **Engage:** ask questions, provoke discussion, *question everything*
3. **Listen:** keep track of what the instructor emphasizes
4. **Keep up:** follow up on things you don't understand
5. **Collaborate:** teaching = max{learning}
6. **Use LLMs smartly:** beware the false sense of 'learning'

References

Bibliography

Wooldridge, Jeffrey M. 2025. "Introductory Econometrics: A Modern Approach."